

USPD S-5-68A
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SUPERSEDING
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UNIFORMS SECTION

PURCHASE DESCRIPTION

CAPS (DISTINCTIVE FOR U.S. ARMY BAND)

1. SCOPE

1.1 Scope. This purchase description covers the requirement for United States Army Band distinctive caps.

1.2 Classification. The caps shall be furnished in the following classes and sizes as specified. (See 6.2)

Class 1 - Officers
Class 2 - Enlisted men

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

C-F-206	- Felt, Sheet: Cloth, Felt Wool, Pressed
H-S-951	- Synthetic Filaments for Brushes; General Specification for
L-P-375	- Plastic Film Flexible, Vinyl Chloride
L-P-390	- Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium and High Density)
CCC-C-432	- Cloth, Sheeting, Cotton (Unbleached, Bleached, and Dyed)

- MIL-C-368 - Cloth, Satin, Rayon and Cloth, Twill, Rayon
- MIL-C-3727 - Cloth, Wool, Barathes
- MIL-S-3577 - Sweatband, Handwear: Leather
- MIL-B-13466 - Braid, Textile and Lace Veilum, Woven, Textile (for Blue Dress Uniform)
- MIL-C-41831 - Cloth, Nylon, Knitted Raschel
- MIL-L-43541 - Leather, Cattlehide, for Visors and Chinstraps
- MIL-C-43675 - Cloth, Interlining, Cotton Warp and Rayon Filling
- MIL-C-82252 - Cloth, Broadcloth, Wool, and Wool - Synthetic (Mothproofed)

STANDARDS

FEDERAL

- FED-STD-191 - Textile Test Methods
- FED-STD-751 - Stitches, Seams, and Stitching

MILITARY

- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

DRAWINGS - U.S. Army Institute of Heraldry
5-2-20 - Cap, Ornamentation - Army Uniforms

OTHER PUBLICATIONS: Army Regulation 670-1

2.2 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

USPD S-5-68A

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents may also be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from the purchase description may appear in the same in which case the purchase description shall govern.

3.2 Individual orders. When caps are made for individual orders, the purchase description shall govern, except that detailed requirements may be amplified to the extent necessary for good cap making practices.

3.3 Materials.

3.3.1 Basic materials.

3.3.1.1 Crown. The basic material for the crown shall be 14.0 ounce wool barathra dyed Blue 150, conforming to type I of MIL-C-3727.

3.3.1.2 Quarters and Band. The material for the side and front quarters as well as the band shall be 14.5 ounce wool broadcloth conforming to type I, class 1 of MIL-C-82252. The side and front quarters shall be dyed Army Scarlet, Cable No. 65006 and the band shall be dyed Black, Cable No. 65018.

3.3.2 Lining. The lining for crown of cap, covering body band and front stay shall be rayon twill cloth dyed a Blue shade matching Army Blue 150 conforming to class 1 of MIL-C-368.

3.3.3 Braid.

3.3.3.1 Band. The braid for the band shall be 1-3/4 inches wide and shall conform to type II, class 2, Style B of MIL-B-13466. The center stripe shall be a Blue shade to match Blue 150.

3.3.3.2 Cord for crown seam. The cord edge braid to be used on the crown joining seam shall conform to type I, class 2 of MIL-B-13466.

3.3.4 Bias tape. The bias tape for the crown seam shall be made from cotton sheeting conforming to type VII or VIII, class 3 of CCC-C-432, except that the sheeting shall contain not less than 6.0 percent starch and protein content including chloroform-soluble and water-soluble material; the colorfastness requirements for light and combined laundering and bleaching shall not apply; the seam efficiency test shall not apply and the shrinkage shall not exceed 6.0 percent in the warp or filling. The sheeting shall be dyed Blue to match Blue 150 or the bias tape may be made from lining material specified in 3.3.2.

3.3.5 Crown protector and binding. The crown protector material for identification card holder and binding shall be neutral color, waterproof, flexible film, 0.004 ± 0.0004 inch thick, conforming to type II, class 1 of L-P-375.

3.3.6 Pressed fiber. The material for the sweatband welt shall be pressed fiber, 1.5 to 2.0 ounces per square foot, 0.015 to 0.020 inch thick, in a natural or black color.

3.3.7 Front quarter stiffener. The material for the front quarter stiffener shall be raschel knitted nylon cloth conforming to MIL-C-41831, except the color shall be optional.

3.3.8 Crown seam tape interlining. The crown seam tape interlining shall be natural or dyed nylon mesh, knitted on a single needle bar raschel and weighing 6.5 ± 0.5 ounces per square yard. The stiffness of the finished cloth shall be as follows:

	Stiffness, load pounds (N)	
	Minimum	Maximum
Two inch specimen; length parallel to wales	0.025	0.045
Two inch specimen; length across or perpendicular to wales	0.040	0.060

3.3.9 Inner body band. The inner body band shall be of high density (linear) polyethylene plastic material meeting the requirements of type III, class H, grades 1 or 2 of L-P-390 except that the low-temperature brittleness, dielectric constant, dissipation factor and thermal-stress cracking resistance properties shall not apply. The bands shall be extruded, stripped, slit or cut $2\frac{1}{4} \pm \frac{1}{16}$ inches wide, with a thickness of 0.030 ± 0.002 inch. At least one side of the material shall have a dull finish.

3.3.9.1 Punching or perforating. The band shall be perforated by punching or other means in a repetitive pattern as approximated by the standard sample with the diameters of the holes as shown by the standard sample. Not less than a 3/8 inch margin at top and bottom of band shall remain unperforated.

3.3.10 Thread. The thread for seaming and stitching the cap shall be in accordance with good commercial practice and in colors to match the basic material.

3.10.1 Colorfastness. The dyed thread shall show "good" fastness to light and wet drycleaning.

3.3.11 Sweatband. The leather for the sweatband shall be 1-3/4 + 1/8 inches wide, dyed a brown shade and shall conform to the requirements of MIL-S-3577.

3.3.11.1 Punching or perforating. The sweatband shall be perforated by punching or other means in a repetitive pattern with the holes measuring 3/32 and 1/16 inch in diameter. There shall be 8 to 9 of the larger holes and 25 to 30 of the smaller holes per 3 inches of linear band and not less than a 3/16 inch margin at top and bottom of band shall remain unperforated.

3.3.12 Visors. The visors shall consist of a top piece of porous material and bottom piece of leather with an interlining of 2.0 ounce (minimum) thick leather to afford required thickness. The visor shall be well bonded and of sufficient stiffness to hold its shape. The inside edge of the visor shall be skived. The outside edge of the visor shall be trimmed and sanded. The visor shall be bounded with 5/8 inch wide gold polyester resin (Mylar or equal), and stitched 1/32 + 1/32 -0 inch from edge of binding strip on top of visor with gold thread. The binding shall measure 1/4 inch on top of visor. The visor shall be molded with heat into a curved shape.

3.3.12.1 Top piece. The top piece of the visors shall be black, homogenous polyurethane reinforced with polyester. The top piece for the class 1 cap only shall be hand embroidered with two arcs of oak leaves in groups of two in accordance with Drawing No. 5-2-20. The embroidered design shall extend from the sides of the visor with a 1/2 inch separation at the front and shall be placed 3/8 inch from front edge of visor. The acorns and stems shall be No. 25 bright bullion; the midribs No. 3 jaceron; and the leaves, acorn tips and cups No. 26 matte finish. The gold wire shall assay not less than 990 silver base with not less than 2 percent gold. A good commercial cotton thread shall be used for stitching the ornamentation. The gold bullion ornamentation shall be lacquered with a minimum of three coats of clear transparent lacquer to protect it from tarnishing. When specified, gold colored rayon or nylon thread may be used for the embroidery.

3.3.12.2 Bottom pieces. The bottom piece of the visor shall be 3.0 to 4.0 leather conforming to type I of MIL-L-43541.

3.3.13 Front stay. The front stay shall consist of a commercial twin wire tape stiffener, one ply of undyed stiff crinoline and one ply of cotton warp and rayon filling cloth or cotton warp and monofilament nylon filling cloth cut to pattern measurements. The stay shall be covered with rayon lining material specified in 3.3.2. The stay components shall meet the following requirements.

3.3.13.1 Twin wire tape stiffener. The twin wire tape shall measure 1 inch wide. The steel wire used in the tape shall be hard rolled, cold finished carbon steel, galvanized, flat wire, and shall be 0.016 ± 0.002 inch thick by 0.090 ± 0.004 inch wide.

3.3.13.2 Crinoline. The crinoline for front stay shall be an undyed commercial crinoline, weighing 1.5 to 2.5 ounces per square yard finished, with not less than 25 yarns per inch in the warp and filling.

3.3.13.3 Cotton warp and rayon filling or cotton warp and monofilament nylon filling cloth. The cotton warp and rayon filling cloth for front stay shall conform to MIL-C-43675. When alternate cotton warp and monofilament nylon filling cloth is used, the fabric shall conform to the following requirements:

Weight (oz/sq.yd.)	7.8 - 8.5
Threads/inch (min)	
Warp	46
Filling	28
Weave	Plain
Stiffness, load lbs. (N) (min)	
Warp	—
Filling	0.20

3.3.14 Staples. The staples shall be stamped from steel wire 0.025 inch ± 0.0005 inch thick. The wire shall be steel, monel metal, brass or stainless steel. If ordinary carbon steel is used, it shall have a non-corrosive finish capable of withstanding not less than 25 hours of the salt spray test.

3.3.14.1 Staples for securing front welt and ends of stitching thereon. The staples for securing the front welt and ends of stitching thereon shall be flat measuring $1/4 \pm 1/64$ inch wide by $3/8 \pm 1/64$ inch leg, 0.0018 ± 0.0003 inch wide by 0.0054 ± 0.0009 inch thick and shall be finished black.

3.3.15 Crown support. The crown support shall consist of a steel wire inclosed in a nylon grommet conforming to the following requirements.

3.3.15.1 Steel wire. The steel wire shall be spring steel 5/16 + 1/16 inch wide and 0.020 + 0.005 inch thick, cut to required size to fit snugly, without distortion to the cap. The wire shall be completely covered with white ethyl cellulose, cellulose propionate or cellulose acetate propionate coating. The coupling shall be 1 + 1/8 inches long and zinc or cadmium coated to withstand 25 hours of the salt spray test.

3.3.15.2 Nylon grommet. The grommet shall be composed of nylon filling yarns and cotton warp yarns braided to form a grommet, approximately 3/4 inch in diameter. The filling yarns shall be of clear nylon monofilament, 0.012 + 0.005 inch thick in diameter conforming to class N, type II of R-S-951, except the cut length designation shall not apply. The warp yarns shall be 12 + 1/2 count singles carded cotton. The grommet shall be made on a 44 or 48 carrier braider, with carriers braiding in pairs with a 1/1 weave with one 0.012 inch diameter nylon monofilament end per carrier; or a 24 carrier braider with carriers braiding in pairs on a 1/1 weave with two 0.012 inch diameter nylon monofilament ends per carrier. Not less than eight warps of 2 ends each of cotton yarn shall be intermittently spaced around the circumference interlacings. There shall not be less than 8 picks per inch in the relaxed state. The ends of the grommet shall be joined by the use of wood or cork plugs, laminated paper rings, or other suitable material to insure proper closure of the grommet.

3.3.16 Felt. The felt for cushioning the front of the cap shall be felt, sheet, wool, pressed, 3/8 inch thick conforming to type I, classification 9R3 of C-P-206. The felt shall be cut in strips 3/4 + 1/8 inch wide. The felt shall not be dyed in colors.

3.3.17 Eyelets. The insignia eyelet for the front of cap shall be brass, enameled black, of appropriate size and in conformance with good commercial practice.

3.3.17.1 Washers. The washers for backing the insignia eyelet shall be brass and shall fit over and make a secure clinch on the eyelet.

3.3.18 Labels. Each cap shall have a 5/16 x 3/16 inch paper size label of good commercial quality affixed to the sweatband 1-1/2 to 2 inches to the left of the joining seam.

3.3.19 Identification card. The identification card of commercial type shall be included and measure approximately 2-3/4 by 1-1/2 inches and contain the following: "Name _____

3.4 Patterns. A size 7-1/8 pattern is available for guidance in manufacturing and for a basic size from which to grade other sizes. The pattern includes guides for cutting, assembling and directional lines for cutting. Request for pattern should be addressed to the U.S. Army Natick Research, Development and Engineering Center, Uniforms Section, Natick, MA 01760-5019. The supplier may adjust the pattern to suit his methods of manufacturing but shall maintain the military design in all details.

Pattern pieces are clearly marked for seam allowances and the supplier shall follow seam allowances with extreme care. The extra width of seam allowance provided on pattern parts is to provide for maximum alteration of garments.

3.4.1 List of pattern parts. The component parts of the caps shall be cut from material as specified in accordance with the pattern parts indicated.

TABLE I. List of pattern parts

Material	Nomenclature of pattern parts	Cut parts
Wool barathes	Crown	1
Wool broadcloth	Front	1
	Side (quarters)	2
	Welt for visor	1
Rayon lining cloth	Crown	1
	Front stay	1
	Front	1
	Side (quarters)	2
Cotton warp and rayon filling cloth or cotton warp and monofilament nylon filling cloth	Front stay	1
Crinoline	Front stay	1
Raschel knitted nylon cloth	Front stiffener	1
Leather; as specified	Visor (to show size and shape)	1
Polyethylene film 0.0040 inch thick	Crown protector	1

3.5 Construction.

3.5.1 General. The caps shall be of the adopted distinctive design of the United States Army Band. The crown shall measure approximately 10-7/8 inches from front to rear and 10-1/4 inches from side to side, based on a size 7-1/8 cap. The width of the front and side quarters as measured from the band joining seam to the center of the cord braid on crown shall measure approximately as follows: center front seam, 2-3/4 inches; side seams, 2 inches; back seam, 2 inches. The width of the finished band shall be approximately 1-7/8 inches. The cap shall be furnished with a grommet, and a wire crown support which fits snugly inside the crown seam to supply tension. The front of the cap shall have one eyelet centered on the front joining seam with the center of eyelet approximately 1-1/4 inches from crown seam. The inside of the crown shall have a waterproofed material crown protector, cut in one oval shape approximately 9-1/2 by 10 inches when sewn to the crown lining. A piece of waterproofed vinyl material 3-7/8 by 2 inches shall be fused onto the crown protector at center front. One of the 2 inch sides shall remain unfused to form a pocket for insertion of an identification card. At the crown seam there shall be a 3/32 inch cord braid sewn to finish between the crown, front, and side quarters around the entire cap. The cap shall be equipped with a visor, as specified, the width of which, at the upper side of the center front shall measure $2 \pm 1/16$ inches.

3.5.2 Linings. The crown, front and side pieces and inner body band shall be fully lined with the blue 150 lining.

3.5.3 Innerbody band. The innerbody band shall be made in accordance with good commercial practice consisting of material specified in 3.3.9 and overlapped 3/4 to 1-1/4 inches and stapled with two staples or securely stitched. The top and bottom of the band shall be bound with bias tape specified in 3.3.4.

3.5.4 Band. The band shall be made of a strip of the basic material (see 3.3.1.2) at approximately 2-1/2 inches wide, and the 1-3/4 inch specified braid. The top edge of the braid shall be caught in the seam joining band to side and front quarters, 1/4 inch from top edge of innerbody band. The bottom shall be stitched 1/16 inch from bottom edge of braid, through basic material and innerbody band. Braid shall be so positioned on the basic material that when bottom band is pulled over the bottom edge of the innerbody band and stitched to the bias tape at bottom of innerbody band, a 1/8 inch of basic material shows below braid at outside bottom of cap.

3.5.5 Visor. The visor with welt shall be centered at front and securely stitched to the cap. Fold a strip of black broadcloth in half, cut in accordance with patterns, over the 3/8 to, -1/16 inch wide reinforcement strip of pressed fiber and stitch 1/4 to 5/16 inch from folded edge. Fold in half a strip of polyethylene film, cut 3/4 $\pm$ 1/16 inch wide, and superimpose on folded basic material (with reinforcement strip inserted) with folded edges even and stitch through all plies 1/8 to 3/16 inch from folded edges to form a welt.

Center the visor and the welt, cut longer than the length of the visor $1/2 + 1/4$ inch on each side, on the inside of the inner body band and stitch together in one operation forming an inside welt for attaching sweatband. The polyethylene strip shall be next to the visor. Raise each end of welt $1/8$ to $3/16$ inch from bottom edge of band and staple through welt and all plies of material including the braid band with the legs of the staple parallel to the stitching in the welt. Ends of staple shall be clinched on outside braid band.

The felt cushion shall be so positioned as to effect a proper cushioning to front of cap. The stitches shall not be clearly visible on outside of band for a distance of 3 feet or more. Hand tack the felt cushion, cut to width as specified and the length of the visor, on inside through inner body band directly above the visor with 5 to 6 stitches with the ends of stitching securely tacked.

3.5.6 Sweatband. The sweatband shall be smooth and flat and hand tacked to inside of cap in such a manner as to give cushioning effect at the front.

3.5.7 Front eyelet. The front eyelet shall be positioned in the center of front piece $1-3/16$ to $1-5/16$ inches above top edge of band measured from center to eyelet. The eyelet shall be inserted through the front piece, lining, and front stay and backed with a brass washer (3.3.17).

3.5.8 Ornamentation.

3.5.8.1 Band. Each cap shall have attached to the band approximately $1/2$ inch in front of each side seam a brass rectangular bar measuring $1-3/4 \times 3/4$ inches. The bar shall have a $1/16$ inch plain brass raised curved border. The inside part of the rectangle shall be mottled brass with four perforations equally spaced in the center lengthwise of the rectangle. The bar shall be slightly curved to the contour of the band and four toggle backed brass stars with a span of $1/4$ inch and a height of $1/16$ inch (see Figure 3), with toggles $1/32$ inch thick shall be inserted in the holes in the bar through the band and innerbody band and toggles shall be spread to hold bar snugly against outer band of cap. When attached, the bottom of the rectangular bar shall be flush with the bottom of the band.

3.5.8.2 Visor. (Class 1 cap only) Visor ornamentation for class 1 caps shall be as specified in AR 670-5.

USPD S-5-68A

3.6 Manufacturing operations requirements. The caps shall be manufactured in accordance with operation requirements specified in table II. The contractor is not required to follow the exact sequence of operations or sub-operations except in the setting of the leather sweatband. Any additional basting or holding stitching, other than those specified, used to facilitate manufacture is permissible provided the threads are removed or do not show in the finished cap.

NO.	TABLE 11 MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ S.C. #	COVER
1.	<u>Cutting.</u> a. Cut cap parts in strict accordance with patterns furnished. b. Cut the outside cap band from wool broadcloth, 3 to 3-1/4 inches wide (see operation 9). c. When more than one sheet, skirt or hood is used for a cap, the parts shall match each other. d. Cut the lining material with the length of the crown piece running in the direction of the warp. Cut the front and side lining pieces across the warp in accordance with patterns furnished. Cut lining material for covering inner body band 2-1/2 to 2-5/8 inches wide on the bias with one end having a selvage edge. e. Cut seam tape material on the bias, $1-5/8 \pm 1/8$ inches wide. Cut the interlining for the crown seam tape $3/4 \pm 1/16$ inch in the warp direction. f. Cut raschel nylon stiffener with the heavy filling lines across the width of the stiffener. g. Cut the crown protector piece and visor in accordance with patterns furnished.						

USPD 5-5-66A

NO.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	ROBBIN/ FOOPER	COVER
2.	<u>Replacement of defective components.</u> During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-657 shall be removed from production and replaced with non-defective and properly matched components.						
3.	<u>Marking.</u> Mark, ticket or bundle all cut parts to insure a uniform shade and proper assembly throughout the cap.						
4.	<u>Make crown seam bias tape.</u> a. The crown seam tape may be cut and stitched or pressed and placed on rolls for machine application. b. (1) Fold sheeting or lining over interlining strip with edges of material abutting or overlapping at center and stitch with two rows of stitching, 3/16 to 1/4 inch gage, through center of tape. or (2) Fold sheeting or lining over interlining strip with edges of material abutting or overlapping at center and press.	301 or 401 or 101		6-8 6-8 6-8	A gold A gold A gold	A gold A gold	

NATICK Form 903
1 Dec 76

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USPD S-5-68A

NO.	TABLE 11 (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	FOOT/COVER	COVER
6.	<u>Make crown.</u> (cont'd)						
	a. Position the film for the identification card holder with the long dimension across front of cap centered ($\pm 1/2$ inch) and $2 \pm 1/2$ inches from front edge of crown protector. Attach to crown protector by heat sealing process or by stitching $1/8$ to $3/16$ inch from edges on three sides only. One short end shall remain open for insertion of card.	301	LSa-1	6-10	A gold	A gold	
	b. Center crown protector on crown lining with identification card holder on outside, and stitch crown protector to crown lining $1/16$ to $3/16$ inch from edge of protector.	301	LSr-2(a)	6-8	A gold	A gold	
	c. Baste crown lining evenly, close to edge of outer crown.	101	SSa-1	4-6	A gold		
7.	<u>Join sides of cap.</u>						
	Finished appearance. The sides of cap shall be carefully joined to avoid twisting or puckering.						
	a. Join front and sides of cap; open seam and stitch each side of seam $1/16$ to $1/32$, - 0 inch. Seams may be pressed open prior to stitching each sides of seams.	301	SSz-3	10-14	A	A	
	b. Join front stiffener to front quarter; stitch front stiffener to top and bottom of front quarter $1/8$ to $3/16$ inch from raw edge with the notches aligned. When stitching bottom cut out portion, the center bottom nylon stiffener shall not be caught in the stitching. Backstitching is not required.	301 or 401 or 101	SSa-1	6-10	A A A	A A A	
	or						

USPD S-5-68A

NO.	TABLE 11 (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOSSIN/ FOOPER	COVER
7.	<u>Join sides of cap.</u> (cont'd) c. Trim top edge of front stiffener 1/4 to 5/16 inch, maintaining curvature. d. Lap a suitable 5/8 to 1 inch bias cut tape (color optional) 3/8 to 3/4 inch on top edge of stiffener and stitch tape to stiffener.	301 or 304	LSa-1	8-12	A	A	
8.	<u>Make front stay (3.2.19).</u> a. Position bottom edge of cotton warp and rayon filling cloth and crinoline (cotton warp, rayon filling cloth face up) on lining at center of lining piece. Fold sides of lining piece over cotton warp, rayon filling cloth toward center. Stitch plies with two vertical rows of stitching centered on stay, with gage stitching 1-1/4 to 1-5/16 inch. NOTE: Black facing shall be against braid in the finished cap. b. Stitch across top of stay 1/2 to 5/8 inch from top edge of stay.	301		6-10	A	A gold	
9.	<u>Make band.</u> a. Cut the outside cap band from basic material wool broadcloth 3 to 3-1/4 inches wide and to length, to fit the size of cap. b. Lap the braid, cut to length, on the cap band with the top edge of braid 1/4 to 5/16 inch from top edge of band. c. Stitch top and bottom edges of braid to cap band with the stitching 1/16 to 1/8 inch from each edge of braid.	301	SSau-2	10-14	A	A	

NATICK Form 903

1 Dec 76

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USPD S-5-68A

NO.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE NUMBER	WOBBLER/ COVER	COVER
9.	<u>Make band. (cont'd)</u> d. Join back seam of band with a 3/8 inch (minimum) seam. Open seam and stitch each side of seam 1/16 + 1/32, - 0 inch. Seam may be pressed open prior to stitching each side of seam.	301	SSz-3	10-14	A	A	
10.	<u>Make lining and attach to quarters.</u> Finished appearance. The front and sides of the lining shall be joined to the front and sides of the cap without twists or puckers. The basting stitches shall not be exposed on outside of the finished cap. a. Join front and sides of the lining pieces. b. Baste the lining sides to the cap sides along top and bottom edges. The basting stitches shall be close to the edge, but shall not be exposed on outside.	301 101	SSa-1 SSa-1	10-14 6-10	A gold A	A gold	
11.	<u>Join band to sides and front and join sides and front to crown.</u> Finished appearance. The band shall be joined to the sides and front and the sides and front joined to the crown of the cap so that the seams are not twisted or puckered. a. Join the top edge of band to the bottom edge of the side with the seam of band aligned with center back seam. The wool broadcloth at top edge of band shall not show on the finished cap, i.e., the top edge of braid shall abut the bottom edge of side(s).	101 or 301 or 500	SSa-1 SSa-1 SSa-1	10-14 10-14 6-8	A A A	A	

USPD 5-5-68A

NO.	TABLE 11 (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	ROBBIN/ LOOPER	COVER
11.	Join band to sides and front and join sides and front to crown. (cont'd)						
	b. Join band to sides and front with cord edge braid situated between both layers with not more than 1/4 inch seam, catching the stiffener and top of front stay in the stitching at center front of cap.	301 or 101	SSad-3(a) SSad-3(a)	10-14	A A	A	
12.	Set eyelets.						
	a. Position one eyelet in center front piece as indicated by mark on pattern (1/8 inch) off mark indicated shall be acceptable and 1-1/2 ± 1/32 inches above top edge of band measured from center of eyelet. The eyelet shall be inserted through the front piece, nylon stiffener and front stay and shall be backed with a brass washer.						
	b. Insert two eyelets in each side of cap, through basic fabric and stiffener on front piece and through basic fabric on side pieces, 7/8 ± 1/8, ± 0 inch from top edge of the band to center of eyelet. The eyelets shall be horizontally positioned, 1 ± 1/8 inches each side of the front and side joining seams measured to center of eyelets. The eyelets may be backed with a brass washer.						

18

USPD 5-5-68A

NO.	TABLE 11 (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	T H R E A D		
					NEEDLE	BOBBIN/ LOOPER	COVER
13.	<u>Set inner body band in cap.</u> Finished appearance. The inner body band shall be so attached in the cap that there shall be no twisting or puckering of the braid band. The front stay shall finish between inner body band and braid band. a. Reverse inner body band and outside of band and position so that the bottom edge of braid shall be $3/16 \pm 1/16$ inch above the bottom edge of the finished cap. b. Stitch top edge of inner body band to seam allowance of braid band and cap $1/4$ to $5/16$ inch from top edge of inner body band. The stitching shall start and finish within $1-1/4$ inches each side of front stay. c. Turn cap and insert the twin wire, cut $3-3/8$ to $3-1/2$ inches long, between rows of stitching, positioned between crinoline and cotton warp and rayon filling cloth of front stay. d. Pull the bottom of band tightly over the bottom edge of the inner body band and stitch $1/16 \pm 1/32$ inch from bottom edge of braid through all plies catching bottom edge of front stay in the stitching. The side edge of the identification-size label shall be caught in the stitching, $1 \pm 1/2$ inches to the right or left of center back seam.	301 or 101	SSa-1 SSa-1	6-8 6-8	A A	A	
		301	OSf-1	8-12	A	A	

NO.	TABLE 11 (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	T H R E A D		
					NEEDLE	ROBBIN/ LOOPER	COVER
14.	<u>Set welt and visor.</u> Finished appearance. The welt and visor shall be uniformly set and the band shall not be distorted or twisted in the setting. a. Fold a strip of black broadcloth (suitable for purpose intended) in half, cut in accordance with patterns furnished, over the $3/8 \pm 0, - 1/16$ inch wide reinforcement strip of pressed fiber and stitch $1/4$ to $5/16$ inch from folded edge. b. Fold in half a strip of polyethylene film, cut $3/4 \pm 1/16$ inch wide, and superimpose on folded basic material (with reinforcement strip inserted) with folded edges even and stitch through all plies $1/8$ to $3/16$ inch from folded edges to form a welt. NOTE: Operations a. and b. may be combined in one operation. c. Center the visor and the welt, cut longer than the length of the visor $1/2 \pm 1/4$ inch on each side, on the inside of the inner body band and stitch together in one operation forming an inside welt for attaching sweatband. The polyethylene strip shall be next to the visor.	301 or 101	BSa-1 BSa-1	6-8 6-8	A A	A	
		301 or 101	SSa-1 SSa-1	6-8 6-8	A A	A A	
		301	LSa-1	6-8	16-4 or 20-4	20-4 20-4	

NO.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	ROBBIN/ LOOPER	COVER
15.	<u>Finish ends of welt.</u> Raise each end of the welt 1/8 to 3/16 inch from bottom edge of band and staple through welt and all plies of material including the braid band with the legs of the staple parallel to the stitching in the welt. Ends of staple shall be clinched on outside braid band.	Hand		16-4 or 20-4			
16	<u>Attach felt cushion.</u> Finished appearance. The felt cushion shall be so positioned as to effect a proper cushioning, to front of cap. The stitches shall not be clearly visible on outside of band for a distance of 3 feet or more. Hand tack the felt cushion, cut to width as specified and the length of the visor, on inside through inner body band directly above the visor with 5 to 6 stitches with the ends of stitching securely tacked.						
17	<u>Make and attach sweatband.</u> Finished appearance. The abutted edges of seam of sweatband shall be even with the center back seam of cap. The sweatband shall finish smooth and flat without distortion to provide a comfortable fit. Neither the sewing stitches nor the edge of the sweatband shall be exposed beyond the lower edge of the cap. a. Apply stamped size marking on inside of the sweatband 2 ± 1/2 inches from end.						

NO.	TABLE 11 (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	NOSE/L LOOPER	COVER
17.	<u>Make and attach sweatband. (cont'd)</u>	304	FSa-1	8-12	A brown	A brown	
	b. Butt ends of the sweatband and join with a row of zigzag stitching, reinforced on the underside with a $5/8 \pm 1/16$ inch strip of cotton fabric. c. Fell sweatband neatly in the cap in such a manner as to give a cushioning effect at front of cap.	Band		7-9	C		
18.	<u>Make and attach removable band.</u> Finished appearance. The removable band shall finish even in width. The band when attached to cap shall be flat and smooth, devoid of fullness, even at crown seam and finish $3/16$ to $1/4$ inch above bottom edge of cap band. a. Join ends with a $1/2$ to $5/8$ inch seam. b. Fit removable band on cap so that the band will be snug without tightness, devoid of wrinkles or puckers.						
19.	<u>Block caps.</u> Finished appearance. The caps shall be uniformly blocked, and shall not be distorted in the blocking process. a. Hand block caps to proper shape over wooden blocks for each specified size.						

USPD S-5-68A

NO.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	POSSIBLE LOOPER	COVER
19.	<u>Block caps. (cont'd)</u> b. The blocking shall be done on a five piece block by thoroughly steaming on the block. The caps shall be thoroughly dried while still on the block in a heated oven.						
20.	<u>Clean cap and attach instruction label.</u> a. Trim all thread ends; remove loose threads, spots, stains and shade tickets without injury to the material. b. Insert the grommet into the crown. The grommet shall be centered on crown seam tape and shall be evenly distributed and fit snugly in the cap. c. Attach cardboard instruction string to one of the chin strap buttons.						

NATICK Form 903

1 Dec 76

EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

USPD 5-3-68A

USPD 5-3-02A

3.8 Measurement of cap. The finished measurements of the caps shall be as specified in table III.

TABLE II. Measurement of caps

	<u>Inches</u>
Height of cap at front (measured at the point where band joins the visor to center of crown seam).	$4\frac{1}{2} \pm \frac{1}{8}$
Width of quarters at side seams (measured from band joining seam to center of seam).	$1\frac{7}{8} \pm \frac{1}{8}$
Width of front piece at center front (measured from band joining seam to crown joining seam).	$2\frac{1}{2} \pm \frac{1}{8}$
Width of sides at backseam (measured from band joining seam to crown joining seam).	$1\frac{7}{8} \pm \frac{1}{8}$
Width of visor (measured on upper side at center front).	$2 \pm \frac{1}{16}$
Width of finished band (measured from front and side seam to bottom edge of band).	$2 \pm \frac{1}{8}$

Crown

Head measurement <u>1/</u>	Size of cap	Length of crown <u>2/</u> (from seam to seam)	Width of crown <u>2/</u> (from seam to seam)
<u>Inches</u>		<u>Inches</u>	<u>Inches</u>
20-1/2	6-1/2	10-1/2	10
20-7/8	6-5/8	10-5/8	10-1/8
21-1/4	6-3/4	10-3/4	10-1/4
21-5/8	6-7/8	10-7/8	10-3/8
22	7	11	10-1/2
22-3/8	7-1/8	11-1/8	10-5/8
22-3/4	7-1/4	11-1/4	10-3/4
23-1/8	7-3/8	11-3/8	10-7/8
23-1/2	7-1/2	11-1/2	11
23-7/8	7-5/8	11-5/8	11-1/8

1/ All head measurements $\pm \frac{1}{8}$, $\pm \frac{1}{4}$ inch

2/ All crown measurements $\pm \frac{3}{16}$ inch

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection. Inspection of the finished cap shall be made by the manufacturer for compliance with this Purchase Description.

4.2 Certificate of compliance. Certificates of compliance for components and materials shall be submitted and the Government reserves the right to inspect such items to determine the validity of the certification.

4.3 Certification. When caps are purchased individually or in lots, the supplier shall certify to the purchaser that the items have been made in accordance with the quality of material and components established by this purchase description and shall include a label incorporating the certificate number assigned by the U.S. Army Uniforms Section. The label shall read as follows:

"This garment is warranted to meet or exceed the standards of Purchase Description USPD S-5-68A and was produced under Certificate No. _____ from basic material and controlled components warranted by the manufacturer to have been produced in accordance with sample under current certificate."

5. PACKAGING

5.1 Packaging and packing. Packaging and packing shall be in accordance with good commercial practice.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The caps are intended for use by personnel of the United States Distinctive Army Band only as a component of the band uniform.

6.2 Acquisition requirements. Acquisition documents should specify the following:

- (a) Title, number, and date of this document
- (b) Class and size required (see 1.2)

6.3 Samples. For access to samples of the end item, address the contracting activity issuing the invitation for bids.

USPD S-5-68A

6.4 Samples, purchase descriptions, and patterns. Request for shade samples, purchase descriptions, and patterns should be addressed to Uniforms Section, U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5019.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Prepared by: Uniforms Section
U.S. Army Natick Research, Development
and Engineering Center
Natick, MA 01760-5019
STRNC-IUU

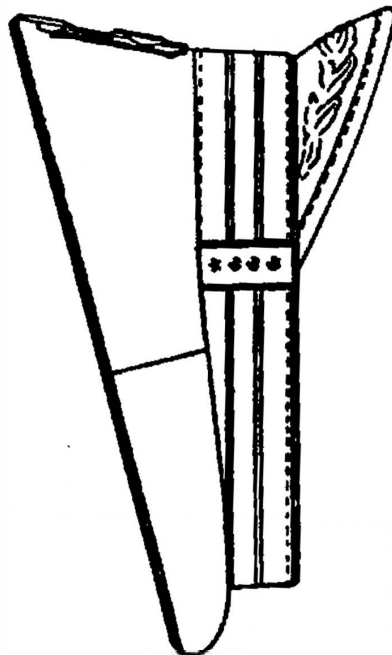
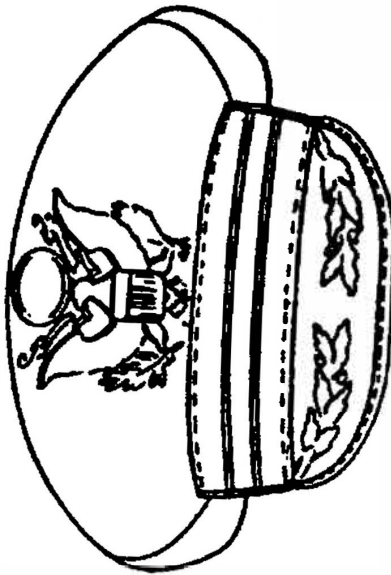
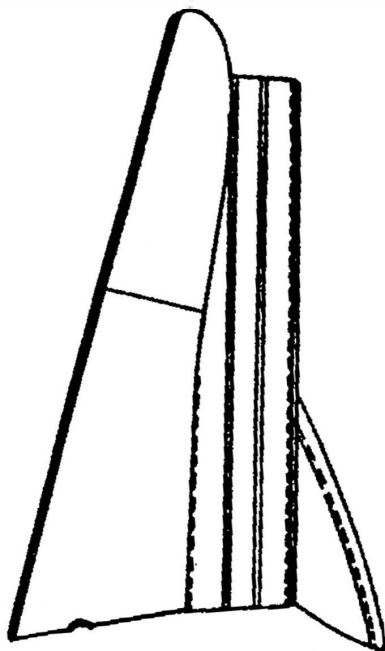
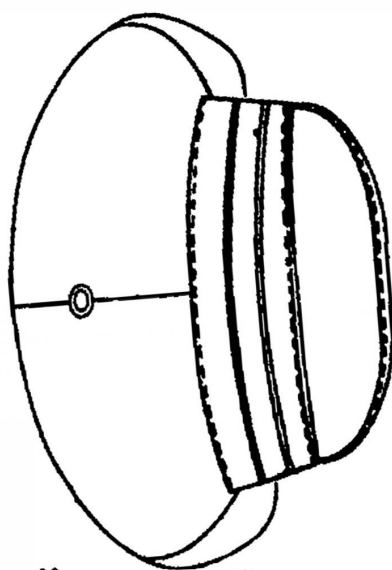


FIGURE 1 OFFICER'S CAP



*FIGURE 2 ENLISTED
MAN'S CAP*



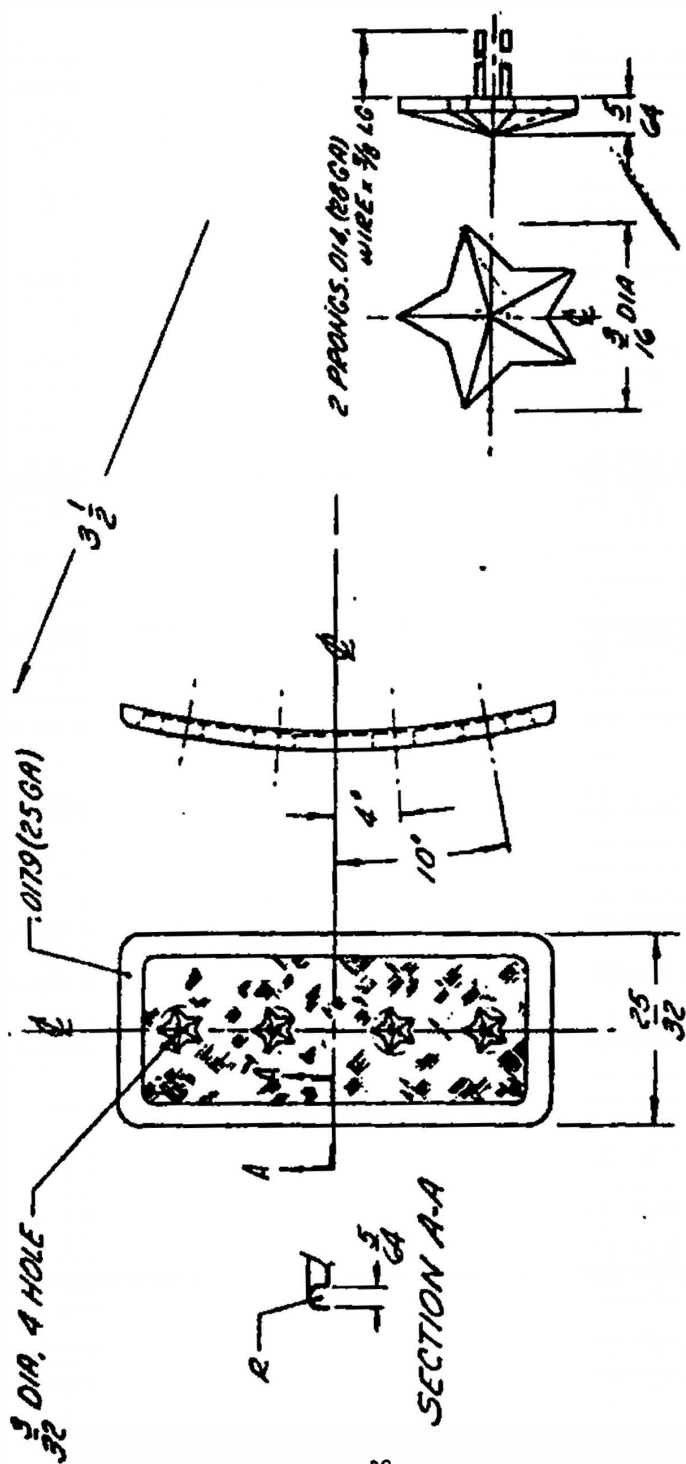


FIGURE 3 BRASS BAR ORNAMENTATION